

D598

Analytics Programming

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An analyst on a 6-person team at a shipping company stores weekly volumes in a Python list: `vols = [120, 135, 150, 160, 175]`. The team also maintains an R version of the same script. What does `vols[3]` return?

- A. [120, 135, 150]
- B. 175
- C. 150
- D. 160

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

In R, `x <- c(120, 135, 150, 160, 175)`. Which expression returns 150?

- A. `x[3:4]`
- B. `x[2]`
- C. `x[3]`
- D. `x[-3]`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A CSV exported from a point-of-sale system in 14 stores loads the price column as text values like '4.50'. The analyst runs `df['price'].sum()` in pandas to get total revenue. The file is 2.3 MB. What is the most likely problem?

- A. The file is too large to fit in memory.
- B. `sum()` always converts text to numbers automatically.
- C. The values are strings, so `sum` joins them together; convert with `pd.to_numeric` first.
- D. The prices need rounding before they can be added.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A marketing analyst has a pandas DataFrame of 2 million transactions and needs a new column equal to $\text{amount} \times 1.08$ for tax. Her laptop has 16 GB of RAM and the data came from a vendor in Ohio. Which approach is most efficient and idiomatic?

- A. Upgrade to 32 GB of RAM first.
- B. `df['with_tax'] = df['amount'] * 1.08`
- C. `df.apply(lambda r: r['amount'] * 1.08, axis=1)`, because apply is always vectorized.
- D. Loop over `df.iterrows()` and append each result to a list.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A Python script in a clinic's analytics unit, which moved from SAS to Python last year, runs `baseline = [10, 20, 30]`, then `scenario = baseline`, then `scenario.append(40)`. The unit reports to the CFO. What is `baseline` now?

- A. `[10, 20, 30, 40]`
- B. `[10, 20, 30]`
- C. An error, because lists cannot change after creation.
- D. `[40]`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Starting from `baseline = [10, 20, 30]`, which assignment lets you append to `scenario` while `baseline` stays `[10, 20, 30]`?

- A. `scenario = baseline`
- B. `scenario = baseline.copy()`
- C. `scenario == baseline`
- D. `scenario = baseline[0]`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An analyst in a 40-person revenue team wants rows of a pandas DataFrame where region is 'West' and sales exceed 5000. The data covers fiscal 2025 and was reviewed by internal audit. Which expression is correct?

- A. `df[(df['region'] == 'West') | (df['sales'] > 5000)]`
- B. `df[df['region'] == 'West' & df['sales'] > 5000]`
- C. `df[(df['region'] == 'West') & (df['sales'] > 5000)]`
- D. `df[(df['region'] == 'West') and (df['sales'] > 5000)]`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A sensor dataset from a water utility has readings `x <- c(4, 6, NA, 10)` in R. The utility serves 80,000 households and was founded in 1952. The analyst runs `mean(x)`. What is returned?

- A. An error, because mean cannot run on vectors
- B. NA
- C. 6.67
- D. 5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

In pandas, `s = pd.Series([4, 6, None, 10])`. What does `s.mean()` return?

- A. About 6.67
- B. NaN
- C. 5.0
- D. An error, because None cannot be in a Series

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A script written by a former intern at a 200-bed hospital runs: `total = 0; for i in range(1, 5): total += i`. The script is stored on a shared drive. What is total after the loop?

- A. 4
- B. 15
- C. 5
- D. 10

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A sales DataFrame has columns rep, region, and amount. The data was pulled from a CRM the company adopted in 2023. The VP wants total amount for each region. Which expression gives it?

- A. df.groupby('region')['amount'].sum()
- B. df['amount'].sum()
- C. df.groupby('rep')['amount'].sum()
- D. df.groupby('region')['amount'].count()

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Which dplyr pipeline returns the average amount for each region, one row per region?

- A. df %>% mutate(avg = mean(amount))
- B. df %>% summarise(avg = mean(amount)) %>% group_by(region)
- C. df %>% group_by(region) %>% filter(amount > mean(amount))
- D. df %>% group_by(region) %>% summarise(avg = mean(amount))

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

An analyst ~~on a 6 person team at a shipping company~~ stores weekly volumes in a Python list: vols = [120, 135, 150, 160, 175]. ~~The team also maintains an R version of the same script.~~ What does vols[3] return?

Target word: value at a Python index **Decides:** Python indexing starts at 0.

A	Adjacent category	That is the slice vols[:3], not a single index.
B	Wrong stem number	The last element, index 4.
C	Confusion set	R-style 1-based counting.
D	Correct	Index 0 is 120, 1 is 135, 2 is 150, 3 is 160.

2. Answer C Concept 1 • transfer

In R, x <- c(120, 135, 150, 160, 175). Which expression returns 150?

Target word: choose the index — reverse of the Python item **Decides:** R indexing starts at 1.

A	Adjacent category	Returns two elements, 150 and 160.
B	Confusion set	Python 0-based habit; in R x[2] is 135.
C	Correct	In R, the third element is x[3].
D	Inverted operation	A negative index in R drops the third element and returns the other four.

3. Answer C Concept 2

A CSV exported from a point-of-sale system ~~in 14 stores~~ loads the price column as text values like '4.50'. The analyst runs df['price'].sum() in pandas to get total revenue. ~~The file is 2.3 MB.~~ What is the most likely problem?

Target word: why the sum is wrong **Decides:** The column is stored as text.

A	Camouflage	2.3 MB is small and does not decide the result.
B	Absolute language	pandas does not silently convert text to numbers when summing.
C	Correct	Summing a text column concatenates strings instead of adding numbers.
D	Dropped element	Rounding does not fix the data type.

4. Answer B Concept 3

A marketing analyst has a pandas DataFrame of 2 million transactions and needs a new column equal to amount × 1.08 for tax. ~~Her laptop has 16 GB of RAM and the data came from a vendor in Ohio.~~ Which approach is most efficient and idiomatic?

Target word: efficient column arithmetic **Decides:** Simple arithmetic on a whole column.

A	Camouflage	Memory is not the issue for this operation.
B	Correct	Vectorized arithmetic runs on the whole column at once.
C	Absolute language	Row-wise apply is a hidden loop, not vectorized.
D	Adjacent category	Works but runs row by row and is far slower.

5. Answer A Concept 4

A Python script in a clinic's analytics unit, ~~which moved from SAS to Python last year,~~ runs `baseline = [10, 20, 30]`, then `scenario = baseline`, then `scenario.append(40)`. ~~The unit reports to the CFO.~~ What is `baseline` now?

Target word: value after changing an alias **Decides:** Both names point to the same list.

A	Correct	<code>scenario</code> is another name for the same list, so the <code>append</code> shows up in <code>baseline</code> .
B	Confusion set	Assumes assignment made a copy.
C	Adjacent category	That describes tuples; lists are mutable.
D	Dropped element	<code>append</code> adds to the list; it does not replace it.

6. Answer B Concept 4 • transfer

Starting from `baseline = [10, 20, 30]`, which assignment lets you `append` to `scenario` while `baseline` stays `[10, 20, 30]`?

Target word: make an independent copy **Decides:** Need a new list object.

A	Dropped element	Creates a second reference, not a copy.
B	Correct	Creates a new list with the same values.
C	Confusion set	<code>==</code> compares; it does not assign.
D	Adjacent category	Takes the single value 10, not a list.

7. Answer C Concept 5

An analyst in a ~~40-person revenue team~~ wants rows of a pandas DataFrame where `region` is 'West' and sales exceed 5000. ~~The data covers fiscal 2025 and was reviewed by internal audit.~~ Which expression is correct?

Target word: correct boolean filter **Decides:** Column conditions need `&` with parentheses.

A	Inverted operation	<code> </code> means or; it returns rows meeting either condition.
B	Dropped element	Without parentheses, <code>&</code> binds first and the comparison fails.
C	Correct	Element-wise <code>&</code> with each condition in parentheses.
D	Confusion set	Python's <code>and</code> cannot combine whole columns and raises an error.

8. Answer B Concept 6

A sensor dataset from a water utility has readings `x <- c(4, 6, NA, 10)` in R. ~~The utility serves 80,000 households and was founded in 1952.~~ The analyst runs `mean(x)`. What is returned?

Target word: R mean with NA **Decides:** R does not drop NA unless told to.

A	Absolute language	<code>mean</code> works on vectors; it returns NA here.
B	Correct	Without <code>na.rm = TRUE</code> , any NA makes the mean NA.
C	Adjacent category	That is <code>mean(x, na.rm = TRUE)</code> , or pandas' default.
D	Confusion set	Treats the missing value as 0: <code>20 / 4</code> .

9. Answer A Concept 6 · transfer

In pandas, `s = pd.Series([4, 6, None, 10])`. What does `s.mean()` return?

Target word: pandas mean with a missing value **Decides:** pandas skips NaN by default.

A	Correct	$(4 + 6 + 10) / 3 = 6.67$; skipna defaults to True.
B	Adjacent category	That is R's default behavior, not pandas'.
C	Confusion set	Treats the missing value as 0.
D	Absolute language	pandas converts None to NaN.

10. Answer D Concept 7

A script ~~written by a former intern at a 200-bed hospital~~ runs: `total = 0; for i in range(1, 5): total += i`. ~~The script is stored on a shared drive.~~ What is total after the loop?

Target word: value of the accumulator **Decides:** `range(1, 5)` yields 1, 2, 3, 4.

A	Adjacent category	The number of passes, not the total.
B	Confusion set	Includes the stop value 5, which range excludes.
C	Wrong stem number	The stop value from the stem.
D	Correct	$1 + 2 + 3 + 4 = 10$.

11. Answer A Concept 8

A sales DataFrame has columns `rep`, `region`, and `amount`. ~~The data was pulled from a CRM the company adopted in 2023.~~ The VP wants total amount for each region. Which expression gives it?

Target word: grouped total **Decides:** Group by region, then sum.

A	Correct	Splits by region and sums each group.
B	Dropped element	One grand total with no grouping.
C	Adjacent category	Totals per rep, not per region.
D	Confusion set	Counts rows per region instead of adding amounts.

12. Answer D Concept 8 · transfer

Which dplyr pipeline returns the average amount for each region, one row per region?

Target word: grouped mean in R **Decides:** `group_by` must come before summarise.

A	Dropped element	Adds the overall mean to every row; no grouping and no collapse.
B	Inverted operation	Summarising first collapses to one row and loses region.
C	Adjacent category	Filters rows above each group's mean rather than returning the mean.
D	Correct	Groups first, then collapses each group to its mean.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	10	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	3	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Indexing: Python vs. R	Python indexes from 0 and slices exclude the end; R indexes from 1, and a negative index in R drops that element.	Carrying one language's indexing habit into the other.	A bracket index on a short list or vector.
2	Data types and conversion	Values read from files may load as text; convert them to numeric types before doing math.	Assuming numbers that look numeric are numeric.	A CSV column that behaves oddly under sum or mean.
3	Vectorization	Whole-column (vectorized) operations in pandas and R are faster and clearer than row-by-row loops.	Writing loops or row-wise apply for simple column arithmetic.	A new column computed from existing columns on a large table.
4	References vs. copies	In Python, assigning a list to a new name creates a second reference to the same object; use <code>.copy()</code> for an independent copy.	Believing <code>b = a</code> makes a separate copy.	A list assigned to another name and then changed.
5	Boolean filtering of data frames	In pandas combine conditions with <code>&</code> and <code> </code> , each wrapped in parentheses; the keywords <code>and/or</code> do not work on columns.	Using <code>and/or</code> or leaving out parentheses.	Two conditions on a DataFrame.
6	Missing values	R's <code>mean</code> returns NA if any value is NA unless <code>na.rm = TRUE</code> ; pandas skips NaN by default.	Assuming both languages treat missing values the same, or treating missing as zero.	NA, NaN, or None inside a numeric vector.
7	Tracing loops	Python's <code>range(a, b)</code> runs from a up to <code>b - 1</code> ; trace the accumulator one pass at a time.	Including the stop value, or confusing iteration count with the result.	A short for loop with a running total.
8	Split-apply-combine	Group by the category you want results for, then apply the aggregate (<code>groupby().sum()</code> in pandas; <code>group_by()</code> then <code>summarise()</code> in dplyr).	Grouping by the wrong column, skipping grouping, or summarising before grouping.	"Per region," "by category," "for each."

Course scope (public description): Analytics Programming builds algorithmic thinking using both the Python and R programming languages. This course builds from the foundations of programming. Learners use libraries and packages to perform common analytics tasks, including acquiring, organizing, and manipulating datasets. The course also presents methods for applying statistical functions and graphical user interfaces to perform basic analysis and to present findings. There is no pre-requisite for this course.